

SIMULATION COMMERCIAL BANK | QUANTITATIVE RISK MANAGEMENT

FORMAL MODEL VALIDATION REPORT

Interest Rate Risk in the Banking Book (IRRBB) & ALM Sensitivity Model

Compliance Framework: Federal Reserve SR 11-7 (Model Risk Management)

Model Risk Tier: High

Date of Validation: July 2026

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Document Control & Version History

Version	Date	Author	Description of Changes
1.0	July 2026	Nicolas Lucero	Initial Validation under SR 11-7 standards.

MODEL VALIDATION REPORT: IRRBB & ALM Sensitivity

Date of Review: July 2026

Model Owner: Corporate Treasury & ALM Desk

Validator: Nicolas Lucero, Independent Quantitative Analyst

1. Executive Summary

- **Model Name:** Asset-Liability Management (ALM) NII & EVE Sensitivity Simulator
 - **Validation Conclusion:** Approved with Conditions
 - **Overall Risk Rating:** High

Validation Abstract

An independent validation was conducted on the bank's Interest Rate Risk in the Banking Book (IRRBB) model. The model calculates both transient Net Interest Income (NII) compression and long-term Economic Value of Equity (EVE) destruction under stochastic rate simulations and regulatory parallel curve shocks. The mathematical framework utilizing a mean-reverting stochastic differential equation (SDE) is conceptually sound. However, the validation identified two critical structural assumptions—the absence of retail deposit betas and the exclusion of prepayment risk—that cause the model to systematically overestimate risk in rising-rate environments.

Key Findings & Conditions for Approval

1. **Condition 1 (Zero-Floor Implementation):** The underlying Vasicek SDE allows for negative interest rates under high-volatility scenarios. A hard-coded 0% floor must be implemented for all US-based retail deposit liability pathways.
2. **Condition 2 (Deposit Beta Integration):** The model currently assumes a 1.0 correlation (100% pass-through) between market rates and liability repricing. Treasury must implement empirical deposit betas before using NII projections for capital allocation.

2. Model Purpose & Use Scope

The model is designed to quantify the balance sheet risk generated by duration mismatch—specifically, the funding of long-term fixed-rate assets with short-term floating-rate liabilities.

- **Approved Use:** To project 12-month Net Interest Income (NII) under Monte Carlo simulated rate paths, and to calculate Economic Value of Equity (EVE) under standard regulatory +200 bps parallel yield curve shocks.
- **Prohibited Use:** Not approved for pricing complex interest rate derivatives (e.g., swaptions) due to the lack of an arbitrage-free framework (such as the Heath-Jarrow-Morton model).

3. Conceptual Soundness

Interest rate dynamics are modeled using the Vasicek framework, mathematically equivalent to an Ornstein-Uhlenbeck physical process.

3.1 Mathematical Framework

The simulation generates forward-looking interest rate paths (r_t) using the following stochastic differential equation:

$$dr_t = a(b - r_t)dt + \sigma dW_t$$

Where:

- a : Speed of mean reversion
- b : Long-term mean interest rate (setpoint)
- σ : Volatility of the rate environment
- dW_t : A standard Wiener process (Brownian motion)

Because this SDE lacks a closed-form analytical solution for discrete time steps, the model correctly implements an **Euler-Maruyama numerical solver** to generate 1,000 Monte Carlo paths.

Validator's Assessment: The Vasicek model is conceptually sound for retail banking ALM. The mean-reverting drift component ($a(b - r_t)$) accurately mimics central bank monetary policy, pulling anomalous rate spikes back to a long-term macroeconomic equilibrium. However, because the volatility term (σ) is independent of the current rate, the model can mathematically produce negative interest rates.

4. Transient Risk: Net Interest Income (NII)

To evaluate the immediate, short-term liquidity impact of interest rate movements, the 1,000 simulated Vasicek paths were applied to a baseline portfolio containing \$100M in fixed-rate assets (5% yield) funded by \$80M in floating-rate liabilities.

4.1 12-Month NII Simulation Results

The liability costs were dynamically recalculated across all 1,000 simulated realities.

Metric	Result (12-Month Horizon)
Asset Yield (Fixed)	\$5,000,000
Expected NII (Mean Path)	\$2,600,000
Best Case NII (Rates drop)	\$3,150,000
99% VaR (Severe Rate Spike)	\$1,120,000

Validator's Conclusion: Under a 99th percentile worst-case scenario (a rapid spike in the Vasicek paths), the bank's NII compresses by over 50%. The model successfully isolates the transient cash-flow risk of duration mismatch.

5. Steady-State Risk: Economic Value of Equity (EVE)

While NII measures short-term cash flow, EVE measures the structural destruction of balance sheet value by calculating the exact Net Present Value (NPV) of all future cash flows.

The validation subjected the portfolio to the regulatory standard **+200 bps parallel shock**.

(Asset Duration: 10 Years | Liability Duration: 1 Year)

5.1 EVE Shock Financial Impact

Metric	Baseline (4% Curve)	Shocked (+200 bps)	Impact
NPV of Assets	\$108,110,000	\$92,640,000	-\$15.47M
NPV of Liabilities	\$79,230,000	\$77,690,000	-\$1.54M
Total Equity (EVE)	\$28,880,000	\$14,950,000	-48.2%

Validator's Conclusion: The +200 bps shock wipes out nearly half of the bank's structural equity. Because the liabilities mature in 1 year, their NPV barely shifts. The 10-year assets, however, suffer massive discounting penalties under the higher rate regime. The math engine correctly computes this duration mismatch.

6. Model Limitations & Critical Assumptions

To satisfy SR 11-7 requirements for "effective challenge," the validation team audited the model's physical assumptions and identified three critical limitations that warp the output:

- Absence of Deposit Betas:** The NII calculation assumes retail deposit rates instantly rise to match the Vasicek market rate. Empirically, retail banks exhibit "sluggishness," passing only ~40% to 60% of rate hikes onto depositors (a deposit beta < 1.0). **Impact:** *The model currently overestimates NII margin compression.*
- Exclusion of Prepayment Risk:** The EVE calculation assumes the 10-year assets will pay out on a fixed, rigid schedule. In reality, if interest rates drop, commercial borrowers will refinance (prepay) their loans early. **Impact:** *The model fails to capture the reinvestment risk inherent in falling-rate environments.*
- Convexity Modeling:** Simple ALM systems often use linear duration approximations ($\Delta EVE \approx -D \times \Delta r$). By building a rigid cash-flow discounting engine, this model successfully captures the non-linear convexity of fixed-income instruments, providing a superior estimate of tail-risk compared to standard linear approximations.

Sign-off:

The model is approved for baseline ALM reporting, provided Treasury overlays empirical deposit betas before executing capital hedges.